

2 Modelling

In the world of computer graphics, the objects in your scene are called models, and represent the objects in your scene. They can range from objects composing a set, to the characters that populate the set. Modelling is similar to sculpting in a sense: as is the case with a good sculptor, a good modeller too, needs to have a good sense of form and volume. While a sculptor works with stone, clay or some other materials, a modeller works virtually, with wireframes and geometry. 3ds Max provides a wealth of tools to shape and sculpt geometry into the shape you desire.

2.1 Understanding geometry types

3ds Max has several ways to represent geometry: meshes, polys, patches, and NURBS. Each type of geometry has its own way of representing a model, and each has its own benefits. Objects can be converted from one type to another, but ultimately, each type of model resolves to triangular polygons when the model is rendered.

2.1.1 Editable mesh

An Editable Mesh is a polygon based object that uses triangular polygons. Editable meshes are useful for creating simple, low-polygonal objects or control meshes for subdivision surfaces. You can convert a NURBS or patch surface to an editable mesh. Editable meshes require little memory and are a natural method of modelling with polygonal objects.

2.1.2 Editable poly

An Editable Poly is a type of deformable object. An editable poly is a polygonal mesh; that is, unlike an editable mesh, it can use more than three-sided polygons. Editable polys are useful in that they avoid invisible edges and have a more-robust set of editing tools. You can convert NURBS surfaces, editable meshes, splines, primitives, and patch surfaces to editable polys.

2.1.3 Editable patch

Editable Patch objects are useful for creating smooth surfaces, and provide very detailed controls for manipulating complex geometry. Unlike a polygonal object, a patch has curvature controls on each vertex to adjust the curvature of the surface.